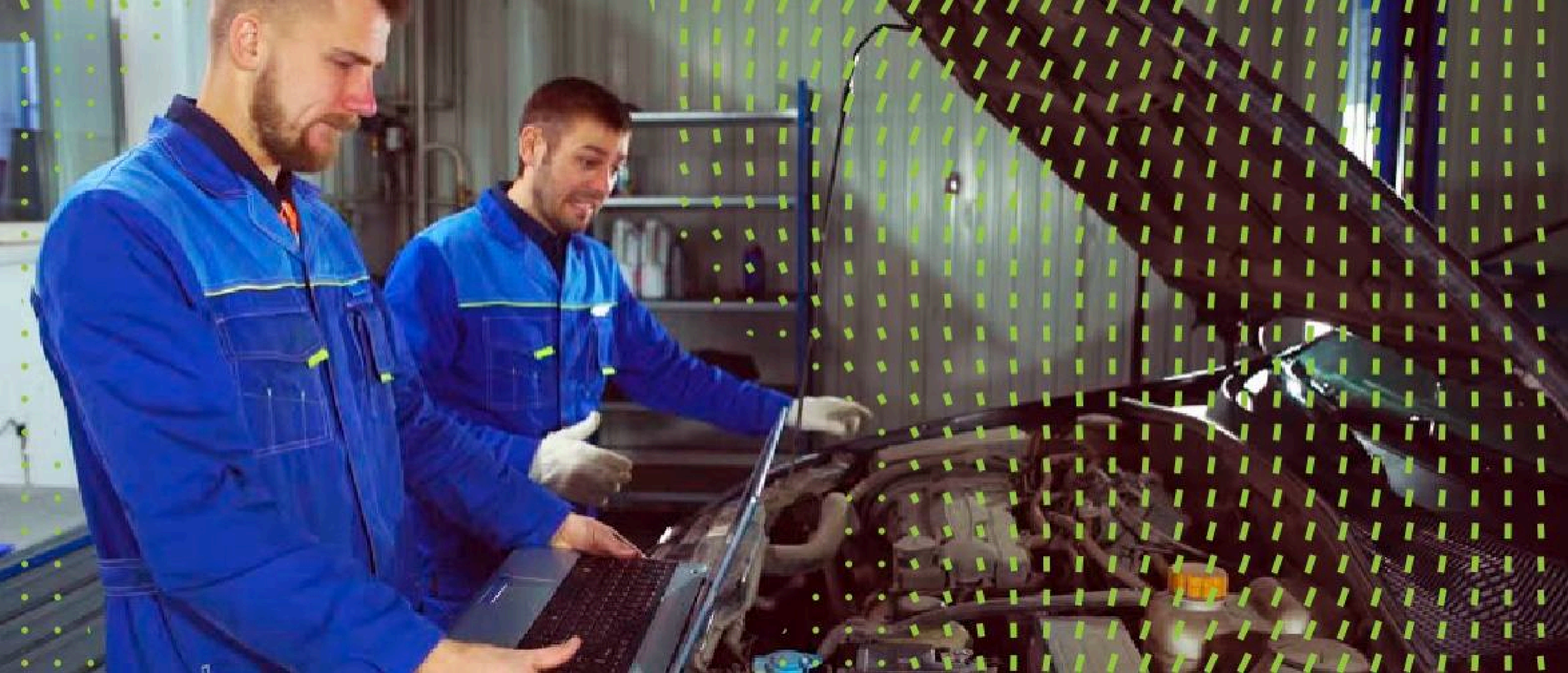




Case Study

Advanced Guided Diagnostics

This case study demonstrates the strategic design approach and UX leadership applied to revamp a legacy technical troubleshooting application, resolving critical pain points in cognitive load through context-first visualization principles.

MY ROLE
UX Designer

PLATFORM
Enterprise Desktop Web App

Business Problem

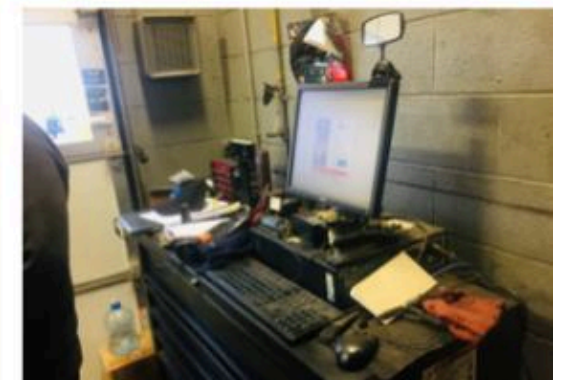
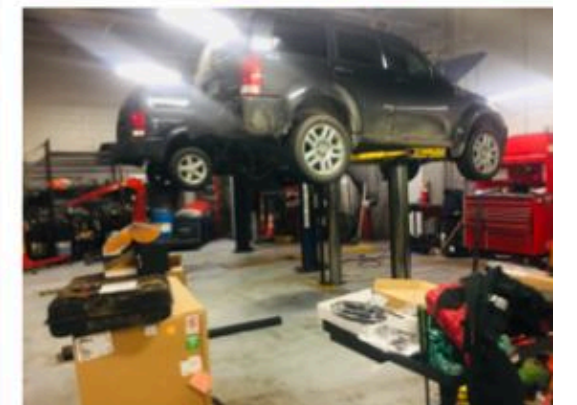
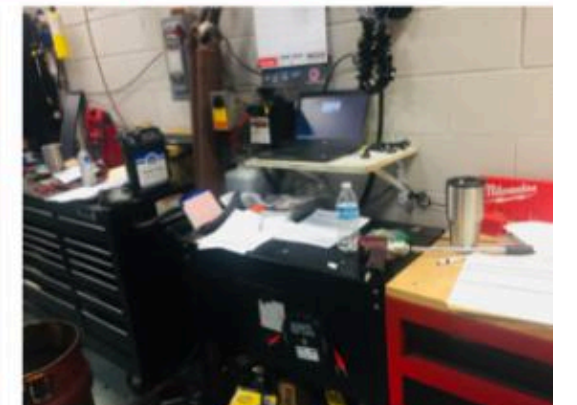
- A way to improve the efficiency of Vehicle Diagnostics
- A system which can guide the service technicians to Perform optimized diagnostic
- A system which can help both novice & expert service technicians
- A platform independent, scalable solution
- A single solution which can work globally across different service centers

Our Approach

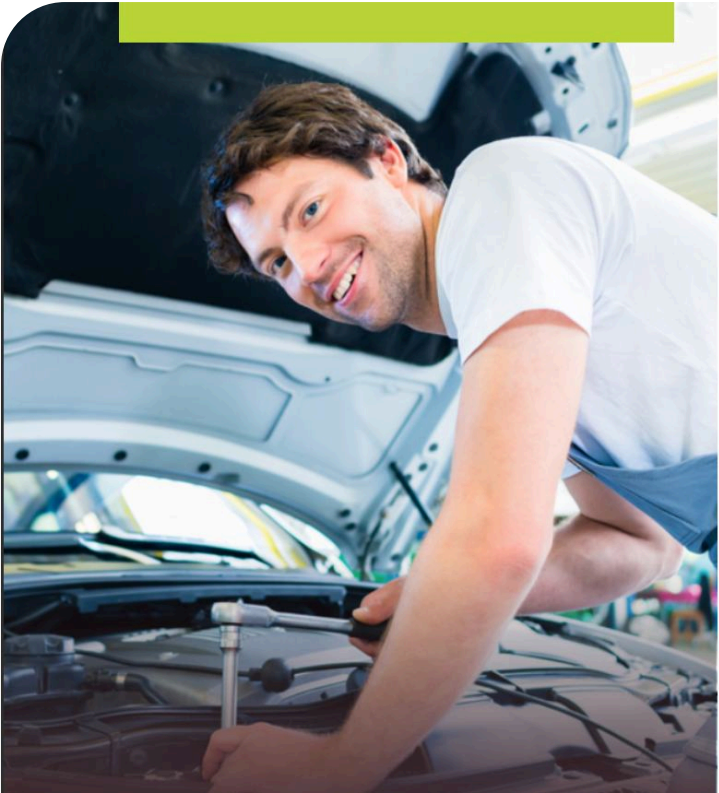
- Apply the principles of user centred design by understanding to service technicians
- User Research – Regular interaction with end service technicians in understanding context, behaviour, needs, motivations, environment variables
- Prototyping a scalable design based on the research with focus on how data for test and repairs are presented
- Usability Testing – Evaluation of design concepts with real users

The Solution

- Guided Diagnostic Application Support
- Recommends the “Test & Repairs” to be executed in vehicle
- Helps decision making by providing Confidence level of each Test/Repair
- A new way to record Symptoms in diagnostics
- Realtime status of Faults & DTC's
- Previous repair history data to help technicians
- Responsive design



User personas



Ray Doe

Junior Service Technician

Age : 26
Experience : 3 Years
Job Title : Junior Service Technician
Status : Single
Education : Mechanical Diploma

ABOUT

Ray is 26 years old junior service technician. He has been working at XYZ dealership for 3 years. He likes to drive sports cars in his free time, indicating his love for automotive elements. Currently, he is responsible for working on scheduled servicing tasks, fitting, oil change, and basic diagnostics. He takes help from the technician experts when required.

“I have used Trace2Fix and love the way it shows the wire color & number”

“It would be helpful if it also can locate the connector”

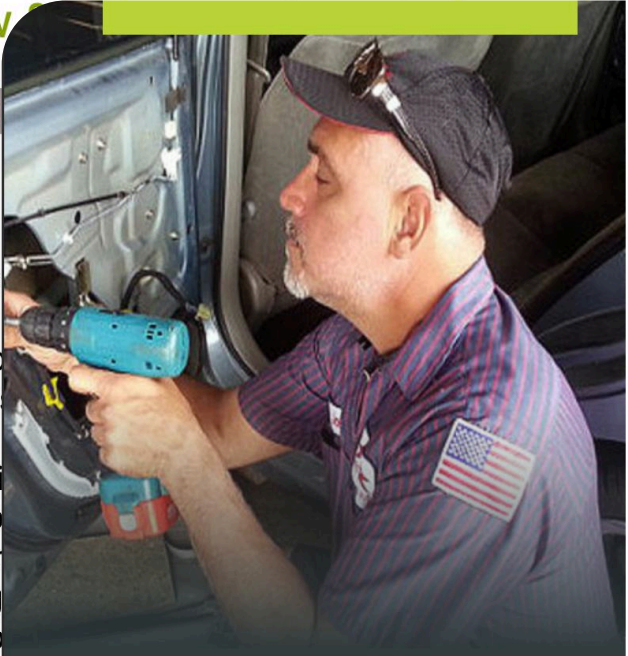
GOALS

- Learn how to solve complex DTCs
- Become an expert of the PCM system
- Able to repair new

Internet Usage

PAIN POINTS

- Dealing with complex iss experienced technicians
- Expert technicians are no
- Need to maintain the per Fixed Visit (FFV) score
- Part names are not consi service manuals, part sto
- Locating the connector in
- Understanding the wiring is problematic since it do



Billy Wilson

Expert Technician

Age : 48
Experience : 26 Years
Job Title : Expert Technician
Status : Married
Education : Basic

ABOUT

Billy is an expert technician with more than 25 years of experience in automotive vehicle diagnostics. He started his journey in his father’s garage in early young age and became passionate about cars. He has gathered practical knowledge while working in various dealerships & OEMs. He is also the expert of PCM ECU and helps other technicians in resolving their problems.

“My FFV score of last week was 92% and this week it is reduced to 89%, this is bad”

“I don’t see Trace2Fix getting open for more vehicle”

GOALS

- Maintain own identity as an expert technician across dealership
- Diagnose the complex issues in a given time frame
- Repair vehicles efficiently to do more jobs each day

Internet Usage

Mobile & Tablet Devices

Computer Literacy

Domain Expertise

PAIN POINTS

- Cars are becoming complex and more issues are at the electrical section
- Diagnosing the fault and identifying root cause takes most of the time
- More workload, hence less time to work on cars
- Expectations are high since due to seniority at dealership
- Always occupied with other technicians for giving guidance, which results in getting own work delayed
- Junior technicians do not have core knowledge of car so need to train them a lot

NEEDS

- Trust is the most important factor for technicians, any new diagnostic tool should provide accurate & detailed information
- Freeze frame data and live parameter readings are important for diagnostics
- Training and helping junior technicians should not impact own work and performance
- Need a way to solve the complex issues of a new vehicle model in less time

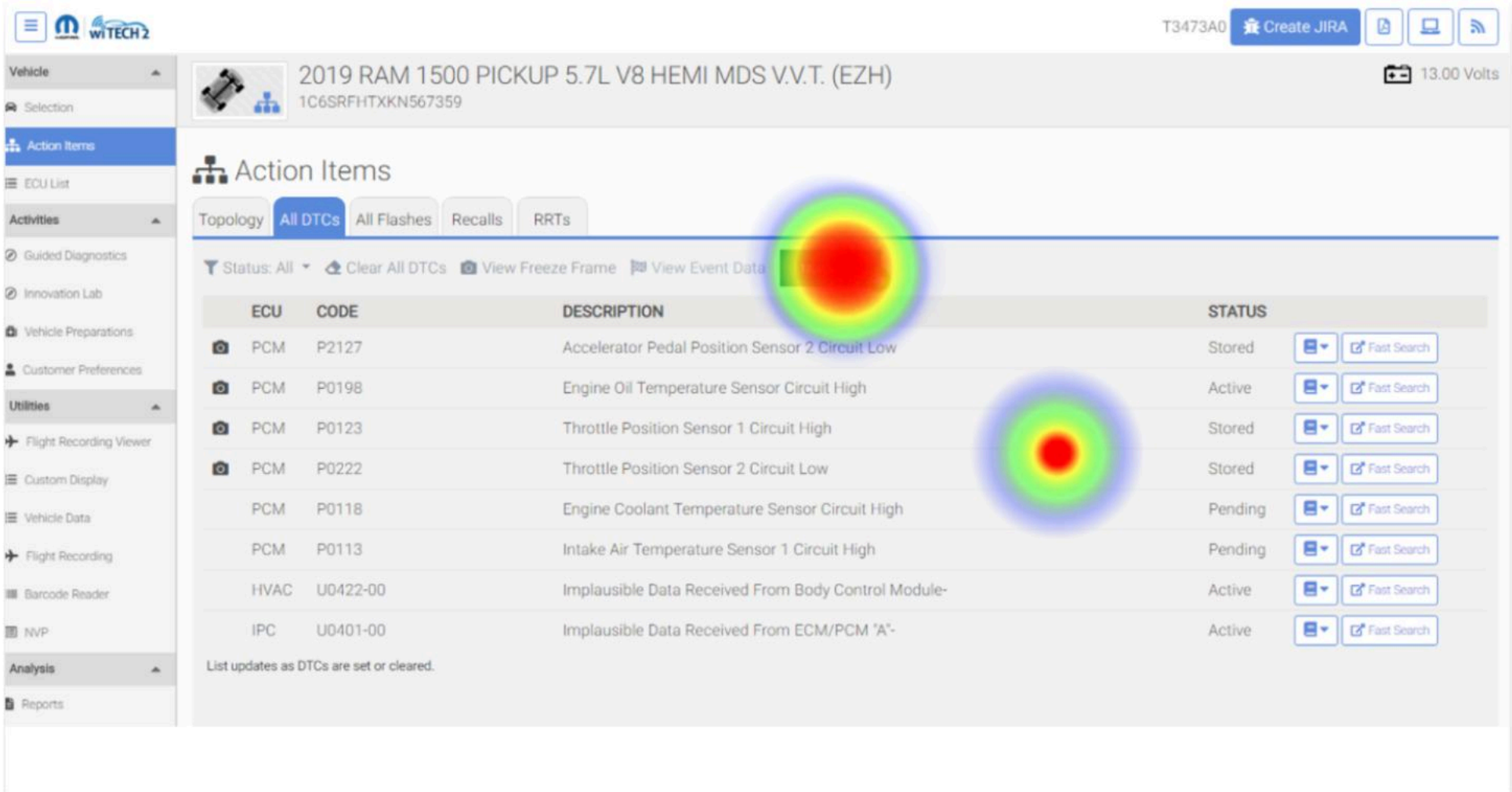
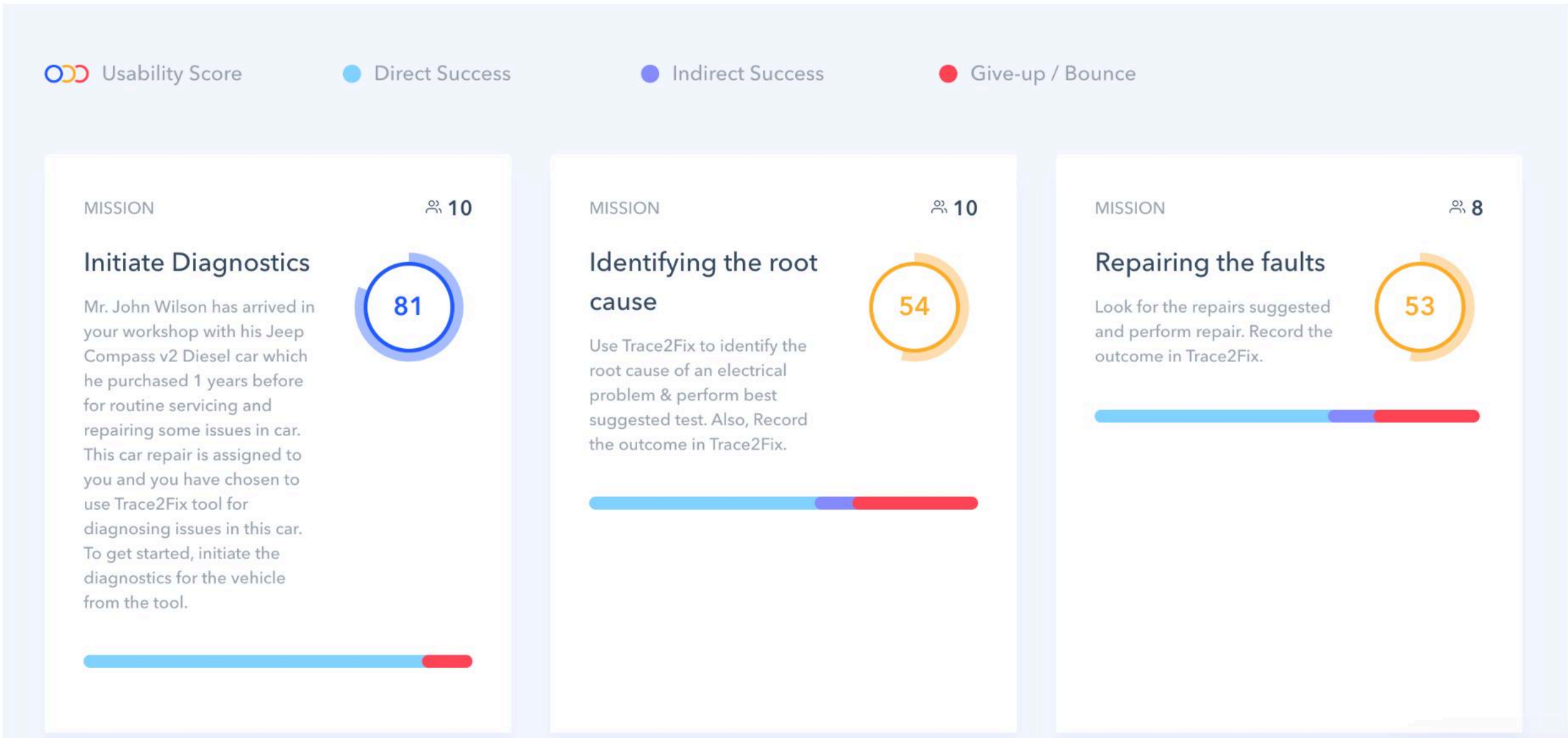
Usability testing & score



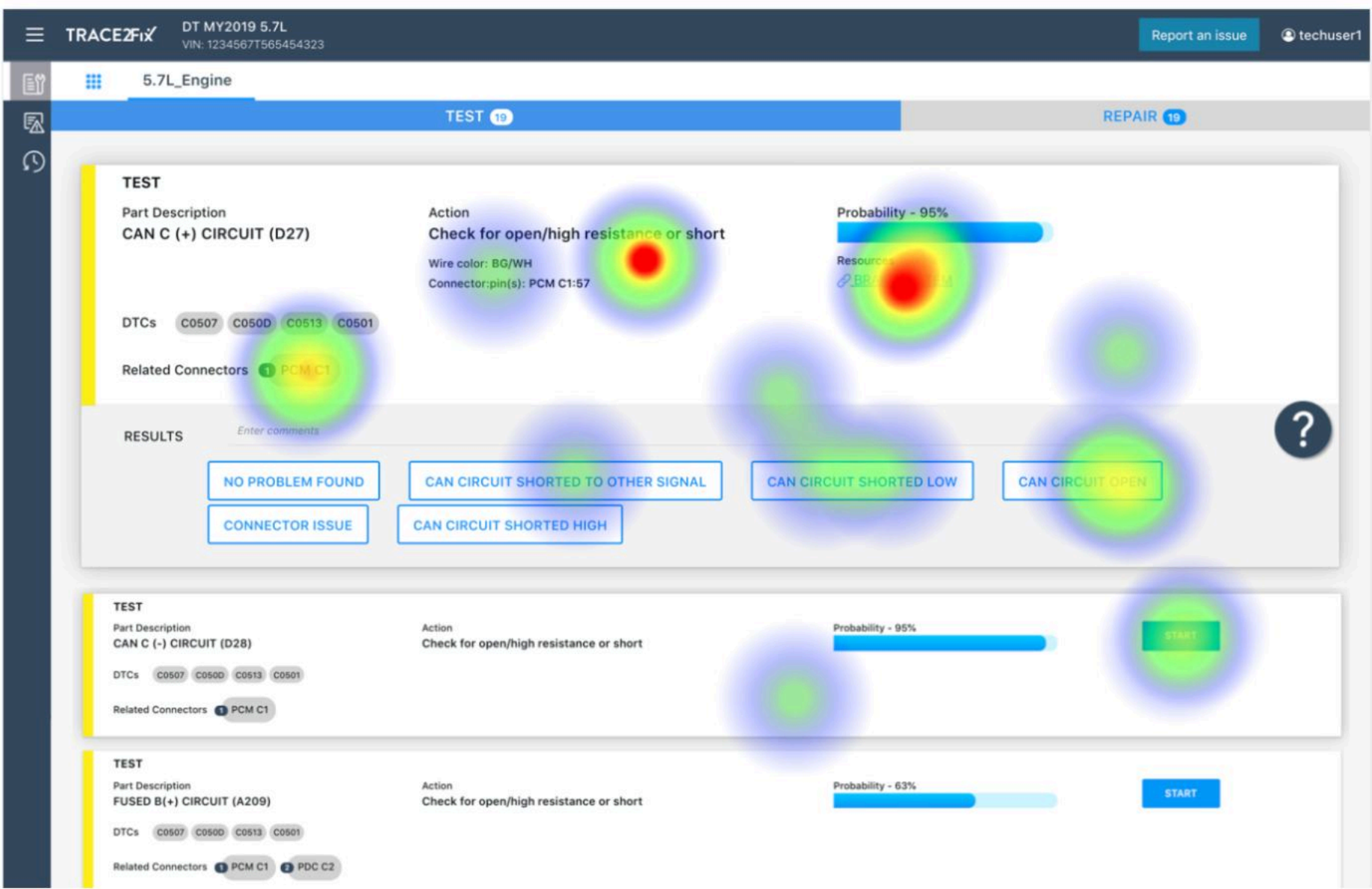
Maze tool for Usability Testing

10

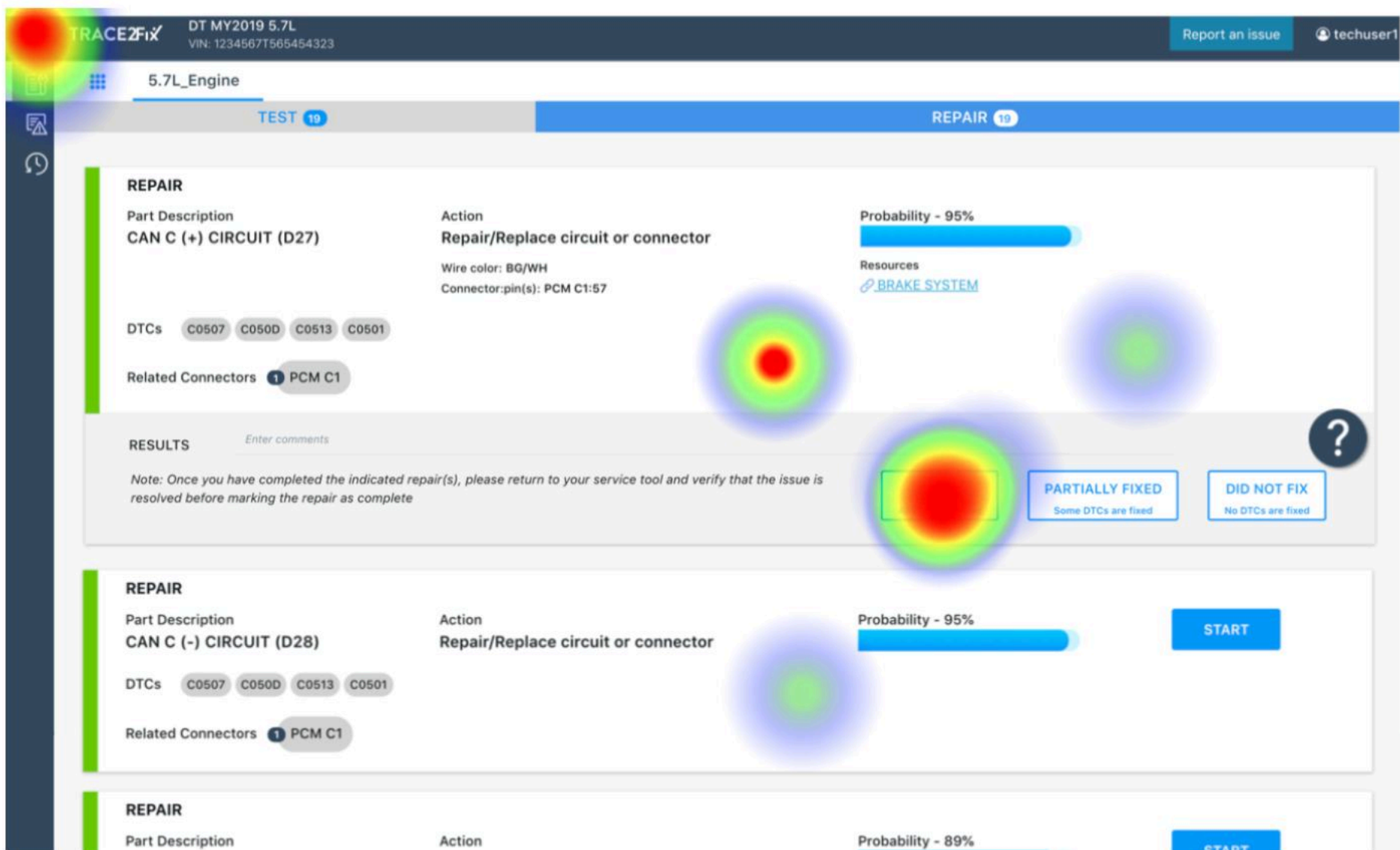
Usability Testing Session



- The majority of technicians were not aware of how to launch Trace2Fix from the WiTech tool, although after providing lead they were able to launch Trace2Fix.
- Some technicians also clicked on a specific DTC to initiate diagnostics.
- Technicians also check freeze frame data and go to the 'Service Library' to seek further steps. Freeze frame data is not available in Trace2Fix application.

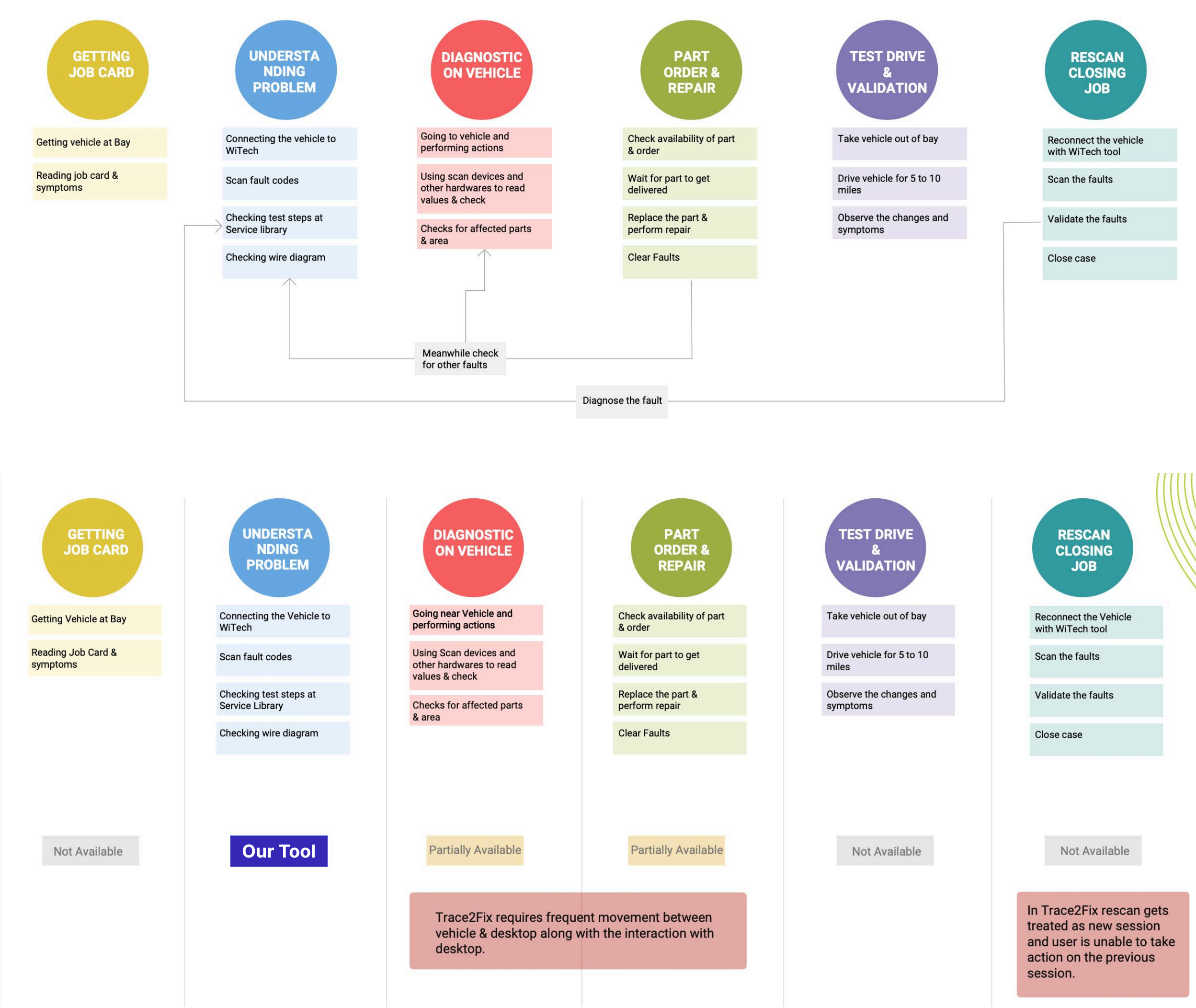


- In first glance, architecture of the screen was not clear to technicians. Majority of technicians did not notice that first test was already expanded.
- Some technicians tried to click on the 'Start' button of the second test card to initiate the first test.
- Initially, technicians were not able to locate the wire diagram link, after providing hint they tried clicking that.
- Technicians also tried to click on 'Connectors' with the assumption that it will open the wiring diagram where they can locate the connector.

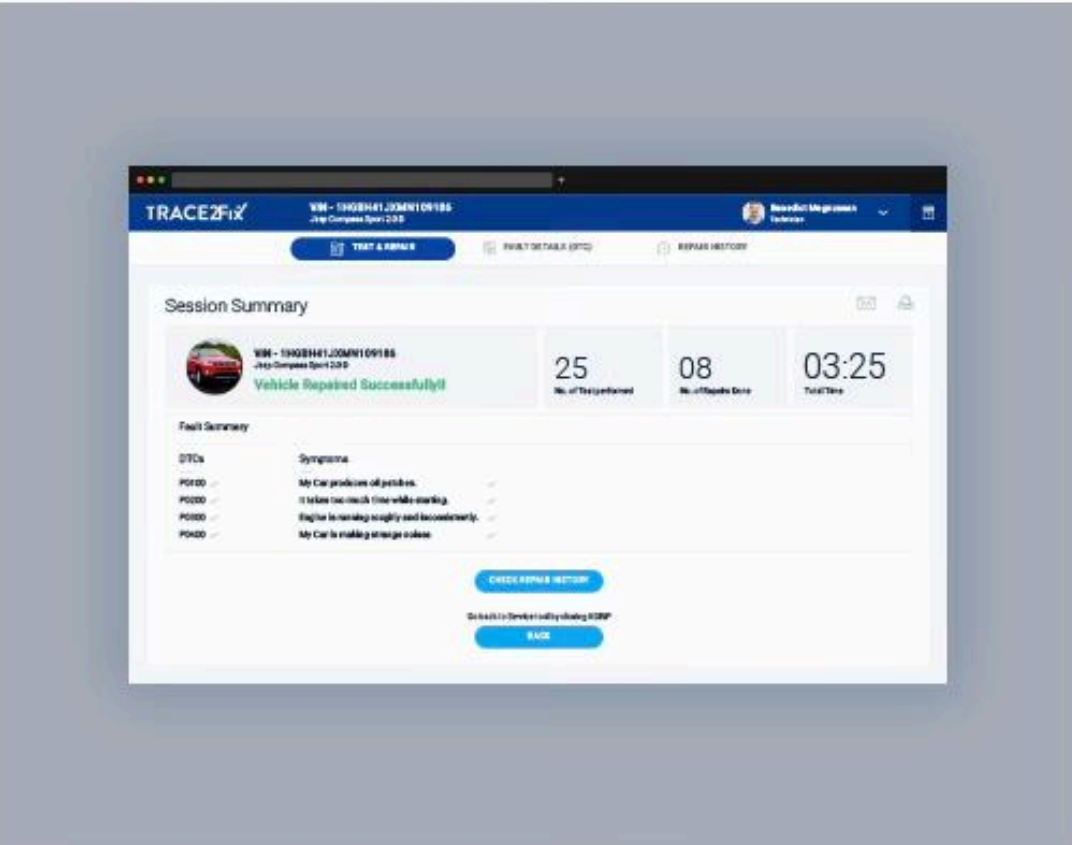
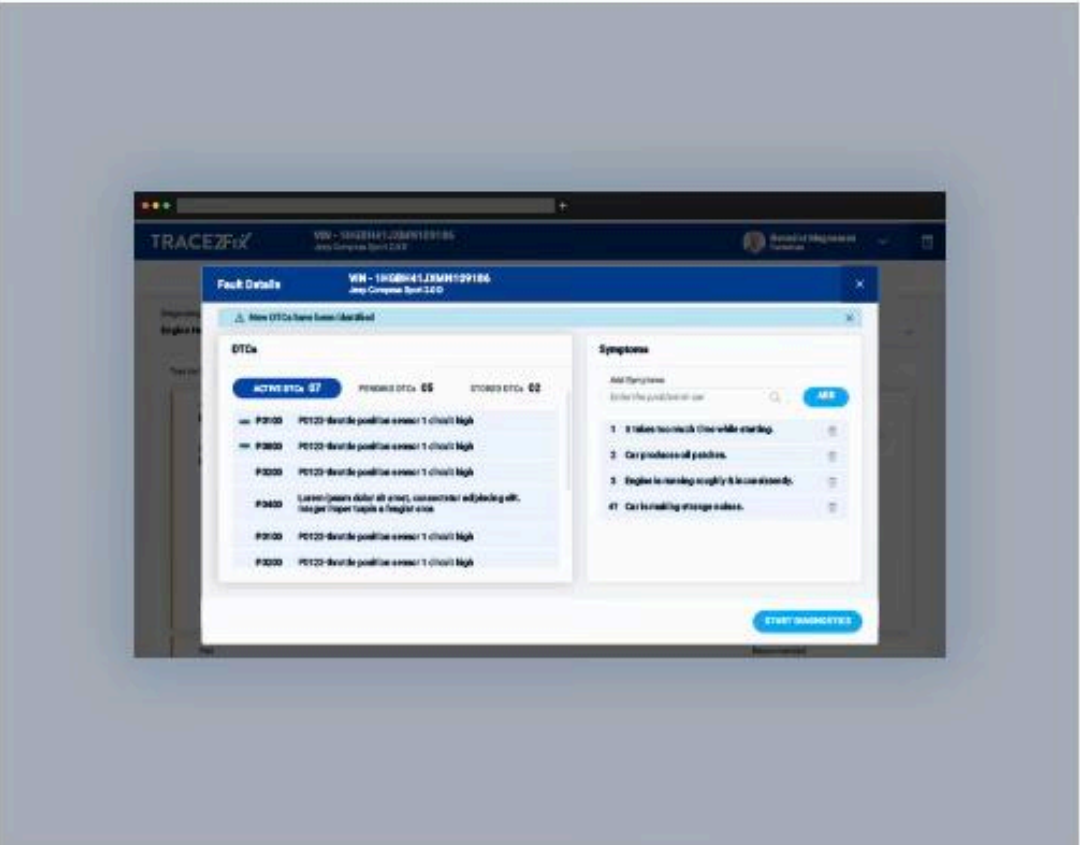
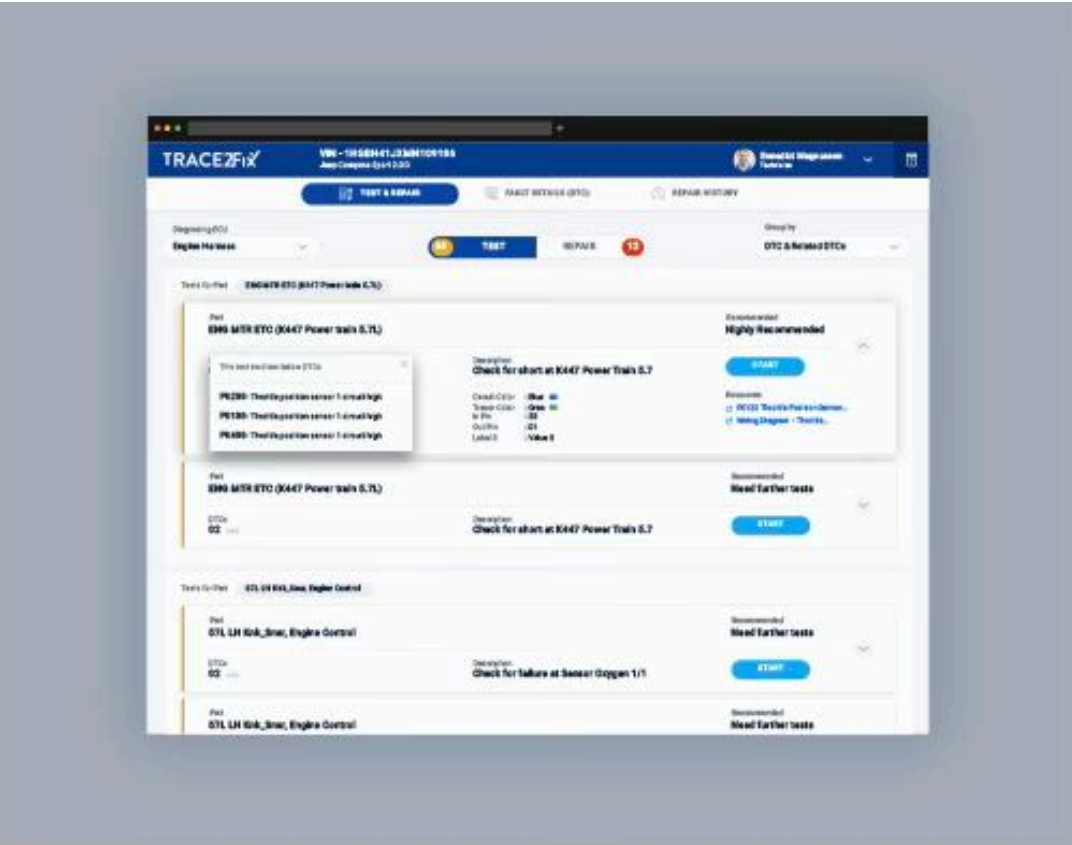
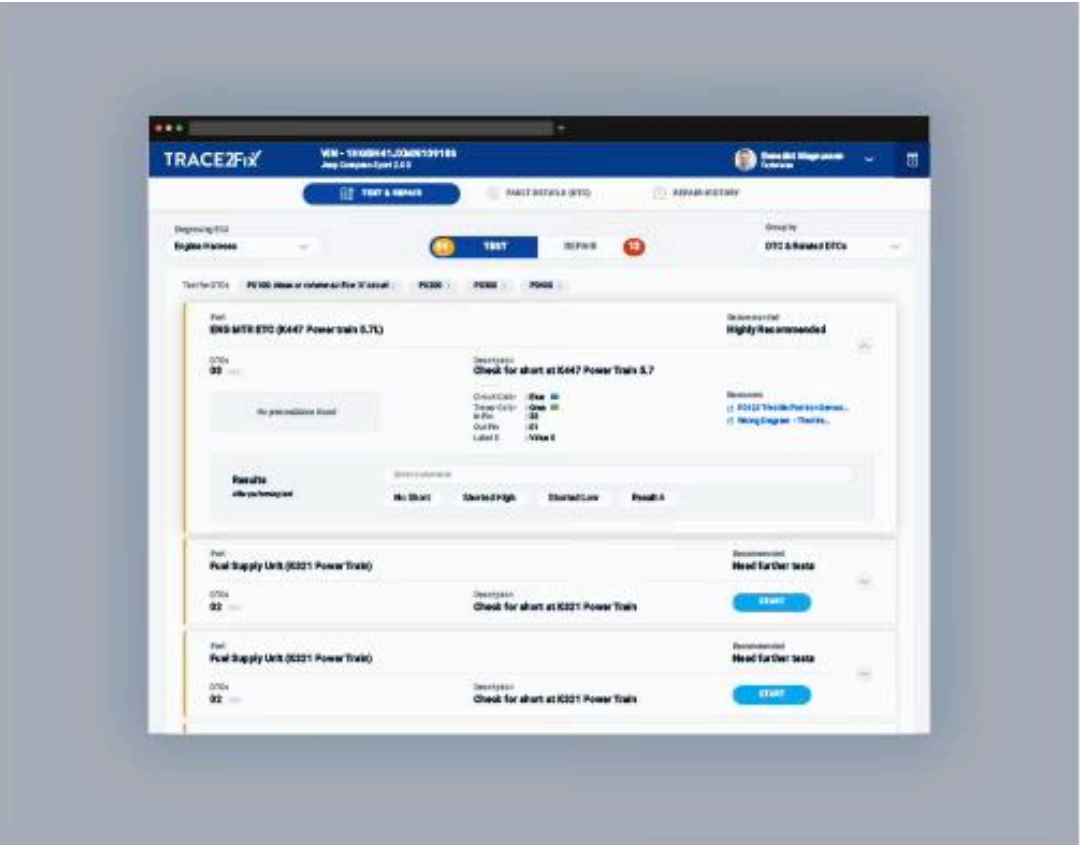
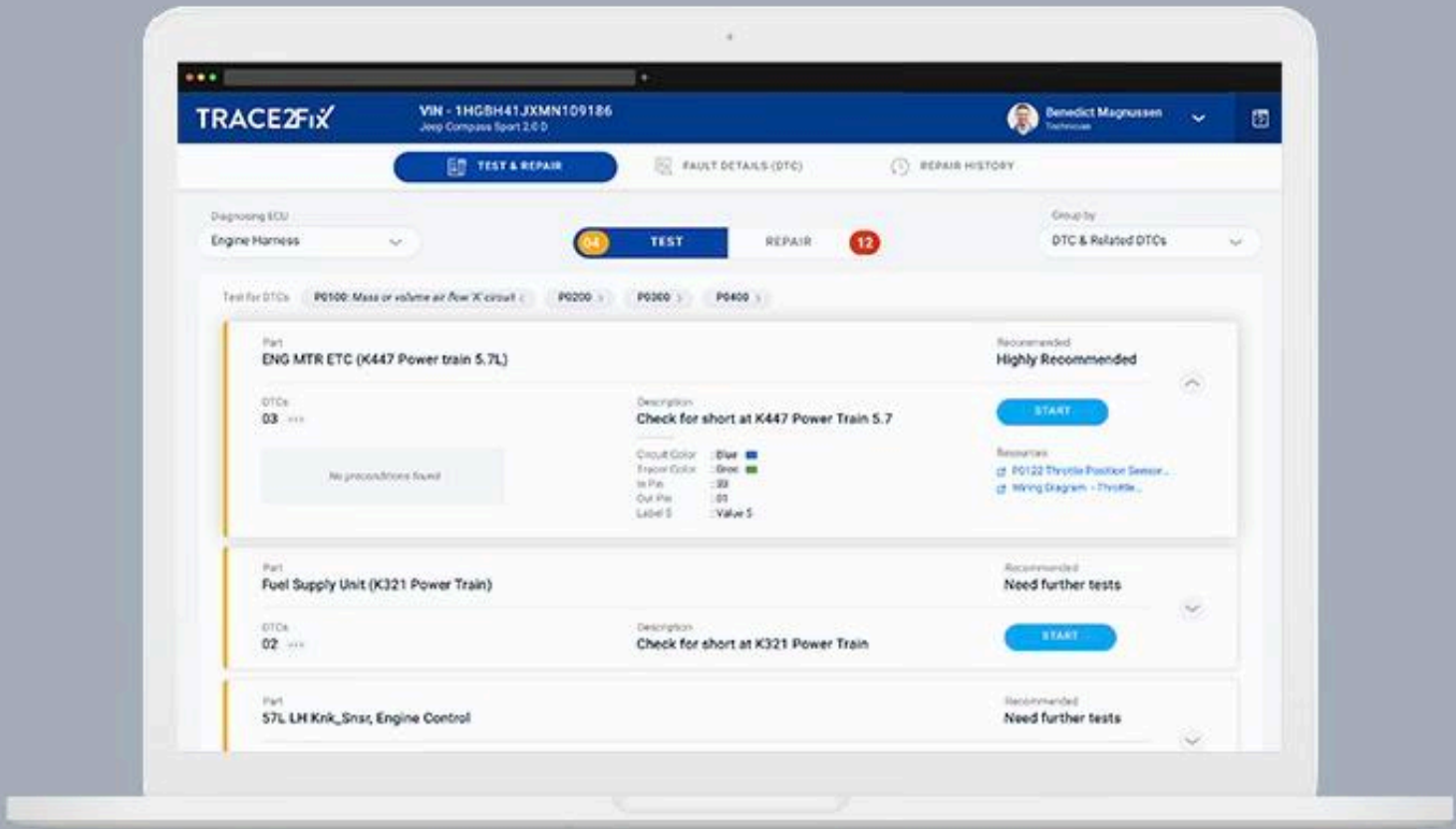


- Some technicians didn't notice the transition from test to repair screen.
- To check the interaction with repair results, technicians were given a scenario of 'test drive and all DTCs are cleared'. With the given scenario technicians were able to click the 'Fixed' button.
- Some technicians tried the menu to go to 'Faults' screen to see DTC status.
- The majority of technicians did not understand the intention of 'Partially Fixed' action.
- While entering the results of given repair, majority of technicians were referring to faults of that repair instead of all faults of vehicle.

Technician Journey



Final Designs



Thank You

